Smart Waste Collection and Route Optimization Platform

AGILE SPRINT PLANNING & USER STORY WORKSHOP

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Software Engineering

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1. Project Vision and Stakeholders

1.1 Project Vision

The rapid growth of urban populations and the increasing volume of daily waste generation have placed enormous pressure on municipal waste management systems across the world. Traditional waste collection processes rely on fixed, calendar-based schedules where collection trucks visit every bin in a zone regardless of whether it is full, half-empty, or barely used. This outdated approach leads to serious inefficiencies: fuel and labor are wasted on collecting near-empty bins, while bins in high-traffic or densely populated areas often overflow before their next scheduled visit, resulting in unhygienic conditions, foul odor, pest infestation, blocked drainage, and repeated citizen complaints. As cities grow smarter and more connected, there is a pressing need to replace this static, inefficient model with an intelligent, responsive, and data-driven system.

The primary vision of this project is to design and develop a Smart Waste Collection and Route Optimization Platform capable of monitoring bin fill levels in real time using IoT sensor technology and generating the most efficient collection routes using optimization algorithms. The proposed system aims to continuously collect bin-level fill data, process it efficiently on a centralized cloud platform, prioritize bins based on urgency, and generate dynamic, optimized collection routes for waste collection vehicles. By applying route optimization techniques such as the Vehicle Routing Problem (VRP) approach, the system can significantly reduce unnecessary vehicle mileage, fuel consumption, and labor hours while ensuring that no bin is left overflowing.

The application is designed to improve operational transparency while drastically reducing manual planning effort. Instead of dispatchers manually deciding collection routes based on guesswork or fixed schedules, fleet managers can rely on an automated system that continuously analyzes real-time bin data and recalculates routes dynamically as conditions change. The proposed solution also provides real-time visualization of truck locations, bin statuses across the city, and historical performance analytics, enabling municipal authorities to make informed, data-backed decisions about fleet allocation, staffing, and infrastructure planning.

Ultimately, the project contributes to building cleaner, smarter, and more sustainable cities by reducing greenhouse gas emissions from waste collection vehicles, minimizing operational costs for municipal corporations, improving citizen satisfaction through timely collection, and supporting long-term environmental and urban planning goals. The platform is envisioned as a scalable solution that can be adopted by any municipal corporation, urban local body, or private waste management contractor seeking to modernize their fleet operations and move toward smarter, more efficient sanitation management.

1.2 Project Objectives

The Smart Waste Collection and Route Optimization Platform has the following objectives:

- To develop an IoT-based bin monitoring system capable of tracking real-time fill levels across all bins in a coverage zone.
- To implement a real-time fleet monitoring dashboard that continuously displays truck locations, route progress, and operational performance.
- To design a bin prioritization module that automatically flags bins nearing full capacity for urgent collection.
- To implement a route optimization engine that generates the shortest, most efficient collection route based on bin urgency, truck capacity, and road network data.
- To develop a dynamic route recalculation system that adjusts active routes in real time when new urgent bins are reported.
- To generate automated citizen notifications and dispatcher alerts whenever collection schedules or route deviations occur, enabling timely response and corrective action.

1.3 Stakeholders

The successful implementation of the platform requires collaboration among multiple stakeholders. Each stakeholder has a specific responsibility in ensuring efficient collection, maintaining service quality, and improving overall municipal waste management operations.

Stakeholder	Role and Responsibility
Municipal Corporation / City Authority	Reviews overall operational performance, monitors budget and service quality, and makes strategic decisions based on collection reports.
Fleet Manager	Supervises collection operations, route planning, truck allocation, and workforce management.
Dispatcher	Monitors live routes, verifies flagged bins, adjusts routes when needed, and coordinates with drivers.
Truck Driver	Operates collection vehicles, follows assigned routes, and reports on-ground issues such as damaged bins.
Maintenance Engineer	Performs preventive and corrective maintenance of trucks and IoT sensor hardware.
System Administrator	Manages user accounts, system security, database maintenance, and software configuration.
IoT Sensor Network	Automatically detects and transmits real-time bin fill-level data using ultrasonic/infrared sensors.
Environmental Compliance Officer	Monitors emission reduction data and ensures the platform supports sustainability compliance reporting.
Citizens / Residents	Receive timely waste collection and optional notifications about scheduled pickups near their location.

1.4 Stakeholder Responsibilities Summary

Stakeholder	Primary Goal
Municipal Corporation	Improve service quality and reduce operational cost
Fleet Manager	Increase collection efficiency
Dispatcher	Ensure smooth, real-time route execution
Truck Driver	Complete assigned collections accurately and on time
Maintenance Engineer	Reduce truck and sensor downtime
System Administrator	Ensure system availability and security
IoT Sensor Network	Detect bin fill levels accurately and in real time
Environmental Compliance Officer	Track and report sustainability metrics
Citizen	Receive timely, reliable waste collection service

2. Epics and User Stories

2.1 Overview

The Smart Waste Collection and Route Optimization Platform is divided into multiple Epics, each representing a major functional area of the system. Every Epic consists of several User Stories that describe the system requirements from the perspective of different stakeholders. These User Stories follow the Agile format: "As a <user>, I want <feature>, so that <benefit>." This approach helps the development team understand user requirements, prioritize features, and deliver the project incrementally through multiple sprints.

Epic 1 – User Authentication and User Management

Description: This Epic focuses on providing secure authentication and authorization mechanisms for all users of the platform. It ensures that only authorized personnel can access the system and perform their respective roles based on assigned permissions.

- US1.1 – Create User Accounts: As a System Administrator, I want to create user accounts so that authorized users can access the platform.
- US1.2 – Secure User Login: As a Dispatcher, I want secure login so that I can access my assigned features.
- US1.3 – Role-Based Access Control: As a System Administrator, I want role-based permissions so that users can access only authorized modules.

Epic 2 – Smart Bin Monitoring

Description: This Epic uses IoT sensor technology to continuously monitor bin fill levels, enabling real-time visibility into waste accumulation across the city.

- US2.1 – Install IoT Fill-Level Sensors: As a System Administrator, I want to install IoT fill-level sensors on waste bins, so that bin fill status can be monitored in real time.
- US2.2 – View Bin Fill Levels on Map: As a Dispatcher, I want to view the current fill level of every bin on a live map, so that I can identify bins that need urgent collection.
- US2.3 – Automatic Bin Flagging: As the system, I want to automatically flag bins above 80% capacity, so that they are prioritized for the next collection route.

Epic 3 – Route Optimization & Planning

Description: This Epic focuses on generating and managing the most efficient collection routes based on real-time bin data, truck capacity, and road network constraints.

- US3.1 – Generate Optimized Routes: As a Fleet Manager, I want the system to generate optimized collection routes based on bin fill levels and truck capacity, so that trucks travel the shortest distance while collecting all priority bins.
- US3.2 – Manual Route Adjustment: As a Dispatcher, I want to manually adjust an auto-generated route before dispatch, so that I can accommodate real-world constraints such as road closures.
- US3.3 – Dynamic Route Recalculation: As a Fleet Manager, I want the system to dynamically recalculate a route when a new urgent bin is reported, so that collection stays efficient without a full manual replan.

Epic 4 – Driver Mobile Application

Description: This Epic provides drivers with a mobile interface for navigation, collection confirmation, and issue reporting during their daily routes.

- US4.1 – Turn-by-Turn Navigation: As a Truck Driver, I want to view my assigned route with turn-by-turn navigation on my mobile device, so that I can complete collections efficiently.
- US4.2 – Mark Bin as Collected: As a Truck Driver, I want to mark a bin as "collected" after emptying it, so that the system updates the bin's status in real time.
- US4.3 – Report Bin Issue: As a Truck Driver, I want to report a damaged or inaccessible bin, so that the maintenance team can be notified immediately.

Epic 5 – Dispatcher / Admin Dashboard

Description: This Epic provides a centralized dashboard that enables real-time monitoring of fleet location, route progress, and resource allocation.

- US5.1 – Real-Time Truck Tracking: As a Dispatcher, I want to view the real-time location of all collection trucks on a map, so that I can monitor fleet progress throughout the day.
- US5.2 – Manage Truck/Driver Assignments: As an Admin, I want to manage truck and driver assignments across zones, so that I can allocate resources efficiently.

Epic 6 – Alerts and Notifications

Description: This Epic provides automatic notifications whenever collection schedules, route deviations, or urgent bin conditions are detected.

- US6.1 – Citizen Collection Notification: As a Citizen, I want to receive a notification when a bin near my location is scheduled for collection, so that I can plan my waste disposal accordingly.
- US6.2 – Route Deviation Alert: As a Dispatcher, I want to receive an alert when a truck deviates significantly from its assigned route, so that I can investigate and respond quickly.

Epic 7 – Reporting and Analytics

Description: This Epic generates centralized operational and sustainability reports to support performance evaluation and municipal decision-making.

- US7.1 – Monthly Performance Report: As a Municipal Manager, I want to view monthly reports on fuel savings, collection efficiency, and route performance, so that I can make data-driven policy decisions.
- US7.2 – Carbon Emission Report: As an Environmental Compliance Officer, I want to view carbon emission reduction estimates from optimized routing, so that I can report on sustainability targets.
- US7.3 – Zone-Wise Performance Comparison: As a Fleet Manager, I want to compare collection performance across different zones, so that underperforming areas can be identified and improved.

Epic 8 – Fleet and Resource Optimization

Description: This Epic focuses on improving overall fleet efficiency by analyzing operational data and recommending optimized scheduling and resource allocation.

- US8.1 – Truck Load Optimization: As a Fleet Manager, I want load-balancing recommendations across trucks, so that no vehicle is overloaded or underutilized.
- US8.2 – Collection Schedule Recommendation: As a Municipal Manager, I want data-driven scheduling recommendations, so that collection frequency matches actual demand per zone.
- US8.3 – Bottleneck Reduction: As a Dispatcher, I want bottleneck reduction recommendations, so that routes are completed within the planned shift time.

3. Acceptance Criteria

3.1 Overview

Acceptance Criteria define the specific conditions that must be met before a User Story is considered complete. They ensure that the developed feature satisfies user requirements, meets quality standards, and functions as expected. Acceptance Criteria also help developers, testers, and stakeholders clearly understand the expected outcome of each User Story.

3.2 Acceptance Criteria for Selected User Stories

US1.1 – Create User Accounts

Acceptance Criteria:

- The System Administrator can create a new user account.
- All mandatory fields must be completed.
- Each user account must have a unique username.
- The system assigns the selected user role successfully.
- A confirmation message is displayed after successful account creation.

US2.2 – View Bin Fill Levels on Map

Acceptance Criteria:

- The dispatcher can log in and open the bin monitoring dashboard.
- All bins are displayed on a live map.
- Bins are color-coded by fill level (Green <50%, Yellow 50–80%, Red >80%).
- Fill-level data refreshes automatically without manual reload.
- Clicking a bin marker shows its exact fill percentage and last updated time.

US3.1 – Generate Optimized Routes

Acceptance Criteria:

- A set of bins has been flagged for collection.
- The dispatcher can request route generation with a single action.
- The system produces an optimized route covering all flagged bins.
- Total travel distance is minimized within truck capacity constraints.
- The route is generated and displayed within 5 seconds.

US3.3 – Dynamic Route Recalculation

Acceptance Criteria:

- A truck is on an active collection route.
- A new bin is flagged as urgent (fill level above 90%).
- The system recalculates the route to include the new bin if truck capacity allows.
- The driver is notified of the updated route immediately.
- The dispatcher dashboard reflects the updated route in real time.

US4.2 – Mark Bin as Collected

Acceptance Criteria:

- A driver has arrived at a bin location.
- The driver can tap “Mark as Collected” within the app.
- The bin status updates to “Empty”.

- The fill level resets to 0% in the system within 10 seconds.
- The updated status is reflected on the dispatcher dashboard in real time.

US5.1 – Real-Time Truck Tracking

Acceptance Criteria:

- Trucks are actively collecting waste.
- The dispatcher can view the fleet dashboard.
- The location of each truck updates on the map at least every 30 seconds.
- Truck status (active, idle, en route) is clearly indicated.
- Historical truck movement can be reviewed for the current day.

US6.1 – Citizen Collection Notification

Acceptance Criteria:

- A citizen has subscribed to alerts for their zone.
- A collection truck is scheduled to arrive within 2 hours.
- The citizen receives a push notification/SMS with the estimated collection time.
- Notification includes the zone name and approximate time window.
- Citizens can opt out of notifications at any time.

US7.1 – Monthly Performance Report

Acceptance Criteria:

- Users can generate monthly performance reports.
- Reports include total distance traveled, fuel saved, and bins collected.
- Reports include average collection time per zone.
- Reports can be downloaded in PDF format.
- Only authorized users (municipal managers, admins) can access reports.

US8.1 – Truck Load Optimization

Acceptance Criteria:

- The system analyzes historical and current load data.
- Load-balancing recommendations are generated per truck.
- Recommendations improve overall fleet utilization.
- Fleet managers can review and approve recommendations.
- Optimization history is stored for future analysis.

3.3 Summary of Acceptance Criteria

User Story	Acceptance Criteria Summary
US1.1	User account created successfully with valid details
US2.2	Bin fill levels displayed accurately, color-coded, in real time
US3.1	Optimized route generated covering all flagged bins
US3.3	Route recalculates dynamically for new urgent bins
US4.2	Bin status updates instantly upon collection confirmation
US5.1	Truck locations tracked and updated in real time

User Story	Acceptance Criteria Summary
US6.1	Citizens receive timely collection notifications
US7.1	Performance reports generated and downloadable
US8.1	Load optimization recommendations provided

4. Product Backlog Prioritization

4.1 Overview

The Product Backlog is a prioritized list of all features, enhancements, and functionalities required to successfully develop the Smart Waste Collection and Route Optimization Platform. The backlog helps the development team focus on delivering the most valuable features first. Priorities are assigned based on business value, user importance, and technical dependency.

4.2 Product Backlog Priority Levels

High Priority

These features are essential for the first release because they provide the core functionality of the system, including secure authentication, real-time bin monitoring, route optimization, and driver collection confirmation.

- User Authentication
- Secure Login
- Role-Based Access
- Install IoT Fill-Level Sensors
- View Bin Fill Levels on Map
- Automatic Bin Flagging
- Generate Optimized Routes
- Turn-by-Turn Navigation
- Mark Bin as Collected

Medium Priority

These features enhance the system by providing additional tracking, dynamic planning, and reporting capabilities.

- Manual Route Adjustment
- Dynamic Route Recalculation
- Report Bin Issue
- Real-Time Truck Tracking
- Manage Truck/Driver Assignments
- Route Deviation Alert
- Monthly Performance Report
- Truck Load Optimization
- Collection Schedule Recommendation

Low Priority

These features improve long-term analysis, citizen engagement, and sustainability reporting but are not mandatory for the first software release.

- Citizen Collection Notification
- Carbon Emission Report
- Zone-Wise Performance Comparison
- Bottleneck Reduction

4.3 Product Backlog Summary

Priority	Number of User Stories
High	9

Priority	Number of User Stories
Medium	9
Low	4
Total	22 User Stories

5. Sprint Backlog Planning

5.1 Overview

The Sprint Backlog is a subset of the Product Backlog selected for implementation during a sprint. It contains the user stories, development tasks, estimated effort, and the scope of work committed for completion. The goal of Sprint 1 is to deliver the core functionalities of the Smart Waste Collection and Route Optimization Platform.

5.2 Sprint Goal

The primary goal of Sprint 1 is to develop the core modules of the Smart Waste Collection and Route Optimization Platform, including secure user authentication, IoT-based bin monitoring, automatic bin flagging, route optimization, and driver collection confirmation. These features provide the foundation for the remaining functionalities to be implemented in future sprints.

5.3 Sprint Duration

Sprint	Duration
Sprint 1	2 Weeks (14 Days)

5.4 Definition of Done (DoD)

A user story is considered “Done” in Sprint 1 when:

- The feature has been developed according to the acceptance criteria.
- The code has been reviewed and merged into the main branch.
- Unit and integration testing have been completed with no critical bugs.
- The feature has been demonstrated to the Product Owner during the Sprint Review.
- Relevant documentation has been updated.

5.5 Sprint 1 Backlog – Stories and Tasks

US1.1 – Create User Accounts

Tasks: Design user schema → Build account creation API → Build admin UI form → Validate unique username → Testing.

US2.1 – Install IoT Fill-Level Sensors

Tasks: Procure sensors → Install on pilot-zone bins → Configure connectivity (LoRaWAN/NB-IoT) → Validate data transmission to backend.

US2.2 – View Bin Fill Levels on Map

Tasks: Design map UI → Integrate map SDK → Build API to fetch bin status → Implement color-coded markers → QA testing.

US2.3 – Automatic Bin Flagging

Tasks: Define threshold logic → Implement backend rule engine → Add flagging indicator on dashboard → Unit tests.

US3.1 – Generate Optimized Routes

Tasks: Select routing algorithm (VRP solver) → Integrate mapping/distance API → Build route generation service → Test with sample bin sets → Performance tuning.

US4.1 – Turn-by-Turn Navigation

Tasks: Design mobile UI → Integrate navigation SDK → Fetch assigned route from backend → Device testing (Android/iOS).

US4.2 – Mark Bin as Collected

Tasks: Build "Mark Collected" action in app → API to update bin status → Real-time sync with dashboard → QA testing.

5.6 Sprint Deliverables

At the end of Sprint 1, the following deliverables will be completed:

- User Registration Module
- Secure Login System with Role-Based Access Control
- IoT Bin Monitoring Module
- Automatic Bin Flagging System
- Route Optimization Engine
- Driver Mobile App with Navigation and Collection Confirmation

6. Story Point Estimation

6.1 Overview

Story Point Estimation is an Agile technique used to estimate the relative effort required to implement each User Story. Instead of measuring the work in hours, Story Points consider the complexity, effort, and uncertainty involved in completing a task. In this project, the Fibonacci sequence (1, 2, 3, 5, 8, 13) is used for estimation because it is a widely accepted Agile estimation method, applied through Planning Poker sessions with the development team.

6.2 Story Point Estimation Scale

Story Points	Complexity	Description
1	Very Low	Very simple task with minimal effort
2	Low	Small enhancement or minor functionality
3	Medium	Standard feature with moderate effort
5	High	Complex feature requiring multiple tasks
8	Very High	Advanced functionality with significant development effort
13	Extremely High	Very complex feature requiring extensive development and testing

6.3 Story Point Estimation Table

User Story ID	User Story	Story Points	Complexity
US1.1	Create User Accounts	3	Medium
US1.2	Secure User Login	2	Low
US1.3	Role-Based Access Control	5	High
US2.1	Install IoT Fill-Level Sensors	5	High
US2.2	View Bin Fill Levels on Map	5	High
US2.3	Automatic Bin Flagging	3	Medium
US3.1	Generate Optimized Routes	8	Very High
US3.2	Manual Route Adjustment	5	High
US3.3	Dynamic Route Recalculation	8	Very High
US4.1	Turn-by-Turn Navigation	5	High
US4.2	Mark Bin as Collected	3	Medium
US4.3	Report Bin Issue	2	Low
US5.1	Real-Time Truck Tracking	5	High
US5.2	Manage Truck/Driver Assignments	3	Medium
US6.1	Citizen Collection Notification	3	Medium
US6.2	Route Deviation Alert	5	High

User Story ID	User Story	Story Points	Complexity
US7.1	Monthly Performance Report	3	Medium
US7.2	Carbon Emission Report	2	Low
US7.3	Zone-Wise Performance Comparison	3	Medium
US8.1	Truck Load Optimization	8	Very High
US8.2	Collection Schedule Recommendation	5	High
US8.3	Bottleneck Reduction	5	High

6.4 Total Story Points

Sprint	User Stories	Total Story Points
Sprint 1	7	29
Remaining Product Backlog	15	68
Overall Project	22	97

6.5 Estimation Justification

The Story Point values were assigned based on the complexity, development effort, technical challenges, and testing requirements of each User Story.

- Low-complexity tasks (2–3 points) include secure login, bin issue reporting, and carbon emission reporting, as they require standard development effort with minimal integration.
- Medium-complexity tasks (3–5 points) include account creation, bin flagging, collection confirmation, and truck/driver assignment, because they involve moderate backend logic and real-time synchronization.
- High-complexity tasks (5 points) include IoT sensor integration, live map monitoring, navigation, and truck tracking, as they require real-time data pipelines and third-party SDK integration.
- Very High-complexity tasks (8 points) include route optimization, dynamic recalculation, and load optimization, as they require optimization algorithms (VRP-based), mapping API integration, and extensive performance testing.

6.6 Benefits of Story Point Estimation

- Helps estimate development effort accurately.
- Improves sprint planning and workload distribution.
- Enables better resource allocation.
- Reduces project risks by identifying complex features early.
- Supports predictable sprint delivery.
- Improves communication among Agile team members.

6.7 Summary

Story Point Estimation provides a structured approach for evaluating the effort required to implement each User Story. By using the Fibonacci estimation technique, the development team can effectively plan sprints, balance workloads, and deliver a high-quality waste collection and route optimization platform within the planned schedule.

7. Sprint Goal and Expected Deliverables

The Smart Waste Collection and Route Optimization Platform provides an effective solution for improving efficiency and sustainability in municipal waste management. By integrating IoT sensors, route optimization algorithms, and real-time data analytics, the system automates bin monitoring, generates efficient collection routes, and supports data-driven decision-making. Compared to the existing fixed-schedule process, the proposed platform improves collection responsiveness, reduces fuel consumption, optimizes fleet utilization, and enhances overall operational efficiency. The use of Agile Scrum as the development methodology ensures flexibility, scalability, and easier maintenance, making the system suitable for future expansion. Overall, the proposed solution helps municipal corporations achieve cleaner cities, lower operational costs, and improved citizen satisfaction.

7.1 Overview

The Sprint Goal defines the primary objective that the Agile team aims to achieve during the sprint. Expected Deliverables are the functional modules, documents, and software components completed at the end of the sprint. These deliverables provide measurable progress toward the successful development of the Smart Waste Collection and Route Optimization Platform.

7.2 Sprint Goal

The goal of Sprint 1 is to develop the core foundation of the Smart Waste Collection and Route Optimization Platform by implementing secure user authentication, IoT-based bin monitoring, automatic bin flagging, route optimization, and driver-side collection confirmation. Successfully completing these features will establish a functional prototype that supports future enhancements and additional sprint development.

7.3 Sprint Objectives

The primary objectives of Sprint 1 are:

- Develop a secure user authentication and authorization system.
- Implement IoT-based real-time bin fill-level monitoring.
- Automatically flag bins nearing full capacity.
- Generate optimized collection routes for flagged bins.
- Enable drivers to navigate routes and confirm collections in real time.

7.4 Expected Deliverables

Deliverable ID	Deliverable	Description
D1	User Authentication Module	User registration, login, and role-based access control.
D2	IoT Bin Monitoring Module	Real-time fill-level data collection and map-based visualization.
D3	Route Optimization Engine	Automated generation of the shortest, most efficient collection route.
D4	Driver Mobile App	Turn-by-turn navigation and bin collection confirmation.
D5	Database Design	Database for storing users, bin data, routes, and reports.
D6	System Integration	Integration of all developed modules into a working prototype.
D7	Testing Report	Unit testing, integration testing, and bug-fixing report.

7.5 Sprint Outcome

At the completion of Sprint 1, the development team will deliver a working prototype of the Smart Waste Collection and Route Optimization Platform. The prototype will include secure authentication, IoT-based bin monitoring, route optimization, and driver-side collection confirmation. These completed features will serve as the foundation for future sprints, where dynamic recalculation, advanced analytics, reporting, and optimization features will be implemented.

7.6 Benefits of Sprint Deliverables

- Establishes a strong foundation for future development.
- Improves collection efficiency through real-time bin monitoring.
- Reduces fuel consumption and unnecessary vehicle mileage.
- Enables real-time visibility into fleet and bin status.
- Enhances decision-making through optimized routing.
- Supports efficient Agile development through incremental feature delivery.

7.7 Sprint Review Summary

At the Sprint Review meeting, stakeholders will verify that:

- All selected User Stories have been completed.
- Acceptance Criteria are satisfied.
- The system functions as expected.
- IoT sensors accurately report bin fill levels.
- The route optimization engine generates efficient routes.
- The driver app correctly confirms collections in real time.
- Feedback is collected for the next sprint.

7.8 Conclusion

The Agile Sprint Planning process provides a structured approach to developing the Smart Waste Collection and Route Optimization Platform. By defining Epics, User Stories, Acceptance Criteria, Product Backlog, Sprint Backlog, Story Point Estimation, Sprint Goals, and Expected Deliverables, the development team can effectively manage project requirements, prioritize tasks, and deliver high-quality software incrementally. This approach improves collaboration, enhances productivity, and ensures the successful implementation of a smart, IoT-driven waste management solution.